Developing Your Geographic Research Project

Causal questions and a feasible analysis plan

Due Wednesday November 11 2026, 11:59 pm | 150 points

Develop a focused causal question for your Quantitative Geography Showcase project, state testable hypotheses, use DAGitty to examine your causal assumptions, and demonstrate that you have usable data. Your final project must include inferential analysis. A purely descriptive or predictive main question does not meet this assignment's requirements.

For this homework, submit a research plan, evidence of data readiness, and a carefully revised DAG that will guide your later regression analysis.

Readings and tools

Read Nick Huntington-Klein's The Effect, Chapters 6 and 7, including Section 6.5 on moderators. Use the class DAGitty demonstration for the adjustment-set task.

Chapter 6 https://theeffectbook.net/ch-CausalDiagrams.html
Chapter 7 https://theeffectbook.net/ch-DrawingCausalDiagrams.html
DAGitty https://dagitty.net/dags.html

1 State your causal question and hypotheses

25 points | About 150–200 words plus hypotheses

- State one question about how a focal exposure or explanatory variable X affects an outcome Y. Identify the study area, population or spatial units, and time period. Specify the contrast of interest, such as an increase in X or a comparison between exposure groups. Focus on the total effect of X on Y for this homework.
- Explain why the question matters and name an audience that could use the answer. Cite at least one credible topic source; you may reuse it to support your diagram.
- State the null hypothesis H_0 and alternative hypothesis H_a in words and, where appropriate, symbols. Describe the population relationship or difference you want to test. You do not need regression notation. Use a two-sided alternative unless you justify a directional alternative in advance.

For example, for a question comparing two exposure groups, H_0 states that their population mean outcomes are equal, while H_a states that they differ. Make your hypotheses specific to your own X, Y, and comparison. Rejecting a statistical null does not by itself demonstrate causation; causal interpretation also depends on study design and assumptions.

Your project begins with a causal question, but your eventual conclusions must reflect what your evidence supports. If important causal assumptions cannot be defended, explain that limit rather than treating an inferential result as proof of a causal effect.

2 Show that you have usable data

30 points | A data inventory, R output, and about 100–150 words

A Document your data

For each core dataset, give its source and link or access route, date accessed, geographic coverage, time period, unit represented by one row, and current number of observations. State which files are already in hand and which are still needed.

Create one compact inventory covering X, Y, likely adjustment variables, the potential moderator, and geographic identifiers. You may also use this inventory as your diagram's variable key.

Concept or node	Actual field and definition	Units and timing	Source and measurement status
X and Y	Field names and what they measure	Coding, units, period	Measured / proxy / must derive / unmeasured
Other variables	One row per variable or concept	Spatial level and period	Dataset or unavailable
Geographic identifier	Join key, coordinates, or geometry	Spatial unit or coordinate system	Boundary or location source

B Provide proof of access in R

Load the core data into R and include the code with readable output showing:

- Dataset dimensions, column names, and a small non-sensitive preview or structural summary, using functions such as dim(), names(), head(), or dplyr::glimpse().
- Show whether X, Y, and your proposed measured adjustment variables are numerical or categorical, and how many values are missing. If either needs to be created from other fields, identify those fields and describe the necessary preparation.
- A usable geographic join key, coordinates, or spatial geometry. For a planned join, identify the boundary source and show its matching key if you have the boundary file; otherwise flag that file as a remaining task.

A download link alone is not proof of data access. For restricted data, submit permitted metadata and aggregate checks rather than confidential records or identifiable locations.

C Assess feasibility

Explain whether the question is manageable using your available data and course methods. Identify two concrete limitations, such as incompatible periods, missing data, small sample size, ecological inference, or uncertain temporal order. State the most important remaining preparation task and a practical fallback if it fails.

You need the core data in hand by submission. Complete cleaning and all joins are not required. If a core variable is unavailable, demonstrate what you do have and propose a narrower causal question that can be investigated with those data.

3 Draw and explain your causal diagram

40 points | One DAGitty diagram, its model code, and a short explanation

A Build the diagram

25 points

- Build one readable DAG in DAGitty, usually around 5–8 nodes. This is a guide rather than a quota. Set X as the exposure and Y as the outcome. Include other plausible causes of X and Y and relevant relationships among those causes.
- Include important unmeasured causes when relevant and mark them unobserved. Do not restrict the diagram to variables available in your data. Explain measurement status in the Section 2 inventory, adding rows for unmeasured nodes as needed.
- Make timing and spatial or population levels clear. If feedback is plausible, distinguish earlier and later measurements so the graph contains no directed cycle. If you combine levels, such as households and neighborhoods, explain that connection.
- Submit a readable image and the complete DAGitty model code, either in an appendix or a plain-text file. Your main DAG should remain a valid graph that DAGitty can use to calculate adjustment sets.

B Explain your assumptions

15 points | About 150–200 words

- Explain three important arrows: what mechanism could produce each relationship, and why is the direction plausible? Support at least two explanations with relevant topic evidence. One source can support more than one arrow and can be reused from Section 1.
- Identify one plausible arrow you left out. Explain the assumption represented by that omission. Then trace one proposed causal route from X to Y and one possible alternative explanation for their association, if present.

Draw arrows from substantive reasoning and plausible timing, rather than from the size or statistical significance of a correlation. A node represents a variable, such as income, rather than a category such as high income.

Useful distinctions

A common cause influences both X and Y. A mediator lies on a causal route from X to Y and helps explain how X affects Y. Identify these roles when relevant, without adding variables simply to include every role.

Submit one carefully revised DAG, rather than separate draft and final diagrams. Review it after the adjustment and moderator tasks, and briefly describe one revision or an uncertain assumption you retained. You do not need to fit an interaction model.

4 Evaluate adjustment and moderation

35 points | DAGitty output and about 250–350 words across A to C

A Choose an adjustment strategy

20 points

Use DAGitty to identify minimal sufficient adjustment sets for the total effect of X on Y. Include its output or a readable screenshot. Minimal means no variable can be removed from that set while preserving its sufficiency under the graph; it does not mean the set is automatically the most practical choice.

- List the reported sets. If there are many, compare two relevant alternatives. If several are available, explain which you prefer using measurement quality, timing, missingness, spatial level, and availability in your actual data. If only one set is available, assess its feasibility.
- An empty adjustment set means no covariate adjustment is required under that graph. This differs from there being no valid measured adjustment set. If no valid measured set exists, explain the obstacle and what additional data or design change might help. You can earn full credit for this conclusion.
- Explain whether your diagram includes a mediator and why you would usually leave it out of the adjustment set when studying the total effect. If it has no mediator, explain this principle briefly without inventing one.
- Use these results to review and finalize your main DAG. If you revise its arrows or nodes, recalculate adjustment sets and submit output matching the final version. Note one revision or remaining uncertain assumption; a second diagram is not required.

B Consider a potential moderator

10 points

Identify a plausible moderator Z: a variable across whose values the effect of X on Y might differ. State where you expect the effect to be stronger, weaker, or different, explain why, and say whether Z is available in your data. Distinguish this proposed effect modification from a mediator explaining how X affects Y.

Show the moderator using a caption or a small annotated copy of your diagram. You may follow the simplified convention in The Effect Section 6.5, including an X × Z interaction node or an annotation pointing to the X-to-Y relationship. This convention is acceptable for this homework; formal DAG notation is not being graded for this illustration. If DAGitty cannot draw the annotation, add it after export or sketch it beside the graph.

Use the ordinary DAG, with Z and its plausible causal arrows if relevant, for adjustment-set calculations. Keep any added interaction node or arrow-to-arrow annotation in the conceptual illustration. You do not need to estimate an interaction or make moderation part of your final analysis.

C State a limit on causal interpretation

5 points

Identify one important assumption your data cannot establish. State how it limits a causal interpretation of the hypothesis test you plan. DAGitty's adjustment output depends on your graph; it does not establish that the graph or its measurements are correct.

5 Plan your analyses and next steps

20 points | A compact table and a brief next step

Create an analysis plan with four columns: output or method; variables and spatial units; purpose; and one relevant choice or limitation. Include these four rows:

- Descriptive summaries for X and Y using methods covered in class, such as means, medians, measures of spread, frequency tables, or distribution plots, as appropriate.
- One proposed map: state the variable and study area it will show, and what geographic pattern it could help you examine.
- One additional exploratory plot or spatial method and what it will help you learn. If using Moran's I or another neighborhood-based method, suggest a definition of neighbors.
- Your future regression analysis in plain language: name Y, the main X, and the measured adjustment variables you propose to include, with the corresponding dataset fields. State how this analysis connects to H_o and H_a. Regression will be taught later; no equation, model type, code, or diagnostic knowledge is required now.

Check that your DAG can guide the final analysis

Confirm that your question, hypotheses, final DAG, adjustment output, and proposed regression variable list use the same X and Y. For every other DAG node, use your inventory to identify its role and actual field or mark it unmeasured. Explain any measured node you will leave out of the regression, such as a mediator, and flag unavailable adjustment variables. Important unmeasured causes should remain on the DAG; flag the limitation rather than silently dropping them. Mark the moderator as a possible extension if you will not analyze it.

Name your two planned Showcase figures and next two preparation tasks. Choose variables from your causal reasoning and data feasibility, rather than statistical significance. Your DAG is a working plan that can be refined after feedback and the regression lessons.

Workload and submission

Allow about 8–10 hours, depending on data readiness. Aim for 900–1,200 words excluding tables, hypotheses, code, output, diagrams, and references. Reuse your variable inventory across sections.

Submit a word file with your final DAGitty code to your Posit folder for HW2. Make sure all your R files/data files are there too. Use Sections 1–5 as headings. Cite sources contributions to approved joint projects.

Grading rubric

Criterion	What earns full credit	Points
1 Question and H_o/H_a	Focused causal question and context 15; aligned H_o and H_a 10.	25
2 Data and feasibility	Documented variables, actual R evidence, realistic limitations and fallback.	30
3A Final DAG	Revised graph, editable code, timing, levels, and measurement key.	25
3B Causal reasoning	Three explained arrows, supporting evidence, omission, and causal routes.	15
4A Adjustment	Output matches final DAG; justified adjustment and revision or uncertainty.	20
4B Moderator	Plausible Z, clear expected modification, illustration, and data availability.	10
4C Causal limits	Specific assumption and its implication for interpreting the planned test.	5
5 Analysis plan	Purposeful summaries and exploration; DAG and data align with regression plan.	20
Total		150